8.8 Period Three Implies Chaos

WORKSHEET

The question

STEP III 2025, Question 2

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What does the graph of $y = 7 - 2|x|$ look like? Where does it cross the line $y = x$? Where does it send the interval $[-7, 7]$?

STAGE 1 The graph of $f \circ f$

Covers: Part (i)(a)

Ask yourself As x goes from -7 to 0, $f(x)$ runs over all of $[-7, 7]$. So what does $f(f(x))$ do on that interval? And from 0 to 7?

STAGE 2 Solving $f(f(x)) = x$

Covers: Part (i)(b) and (c)

Ask yourself Two solutions are the fixed points of f. Why do they also solve $f(f(x)) = x$?

Pause and try For any other solution p, let $q = f(p)$, so $f(q) = p$. Can p and q have the same sign? If not, write two linear equations for p and q and solve. Check there are no solutions outside $[-7, 7]$.

STAGE 3 Arriving at a cycle late

Covers: Part (i)(d)

Ask yourself Compute the first few terms from $a = \frac{28}{5}$. When does it join the 2-cycle?

STAGE 4 Three-cycles

Covers: Part (ii)(a) and (b)

Ask yourself Compute the sequences from $a = 1$ and $a = -\frac{7}{9}$.

Pause and try How many straight pieces does the graph of $\mathrm{f} \circ \mathrm{f} \circ \mathrm{f}$ have on $[-7, 7]$, and how many times can each cross $y = x$? Count the solutions you already know.

STAGE 5 **A late arrival at a 3-cycle**

Covers: Part (ii)(c)

Pause and try Work backwards from the cycle $1 \to 5 \to -3$, as in Part (i)(d). Which numbers does f send to 1? Pick the one not in the cycle, and keep going back.

What's really going on

Cycles of every length

The counting in Stage 4 works for any number of steps. Write f^n for f applied n times. On $[-7,7]$ the graph of f^n has 2^n straight pieces, each running the full height, so $f^n(x) = x$ has exactly 2^n solutions. Some of them have a shorter period d dividing n, but there are at most 2^d of those for each d, and

$$2^n > 2^{n/2} + 2^{n/3} + \cdots + 2^1 \quad \text{(summing over the divisors } d < n \text{ of } n)$$

because all the divisors less than n are at most $n/2$, and $2^1 + 2^2 + \cdots + 2^{\lfloor n/2 \rfloor} < 2^{\lfloor n/2 \rfloor + 1} \leq 2^n$. So for every n there are points whose period is exactly n. This function has cycles of every length.

Why period three forces everything

Li and Yorke's theorem says this is not special to our function: *any* continuous g from an interval to itself with a 3-cycle has cycles of every length. The proof uses nothing more than the intermediate value theorem, and our 3-cycle shows how it goes.

Ask yourself Let $I_1 = [-3,1]$ and $I_2 = [1,5]$, the two intervals between the points of the cycle $1 \to 5 \to -3$. Where does f send each of them?

On I_2, f is decreasing from $f(1) = 5$ to $f(5) = -3$, so $f(I_2) = [-3,5] = I_1 \cup I_2$. On I_1, f goes from $f(-3) = 1$ up to $f(0) = 7$ and down to $f(1) = 5$, so $f(I_1) = [1,7]$, which contains I_2. Notice that only the three points of the cycle were used: any continuous function with a 3-cycle $a \to b \to c$, $a < b < c$, covers its intervals in the same way. We

can summarise it as a diagram of which interval covers which (Figure 8.8.3):

$$I_1 \to I_2, \qquad I_2 \to I_1, \qquad I_2 \to I_2.$$

The theorem then follows from two facts, both consequences of the intermediate value theorem.

- If $g(I) \supseteq I$ then g has a fixed point in I. (Apply the intermediate value theorem to $g(x) - x$.)
- If $g(I) \supseteq J$ then there is a smaller closed interval $I' \subseteq I$ with $g(I') = J$ exactly.

To find a point of period n, follow the loop $I_2 \to I_2 \to \cdots \to I_2 \to I_1 \to I_2$ in the diagram: $n-1$ steps inside I_2 and one step out to I_1 and back. Using the second fact repeatedly, we can shrink I_2 to an interval of starting points whose orbits follow this itinerary, and g^n maps that interval onto all of I_2. The first fact gives a point x with $g^n(x) = x$ that visits I_1 only at step $n-1$. Its period cannot be smaller than n, because then the visit to I_1 would be repeated earlier. (There is one subtlety, since I_1 and I_2 share the point 1; checking that it causes no trouble is an exercise.)

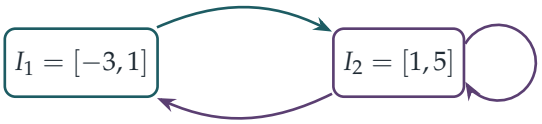


Figure 8.8.3 The covering diagram of a 3-cycle. An arrow from one interval to another means the image of the first contains the second. Every loop in the diagram gives a cycle of the function: the loop $I_2 \to I_2$ gives a fixed point, $I_1 \to I_2 \to I_1$ a 2-cycle, and going round the loop at I_2 $n-1$ times before visiting I_1 gives a cycle of length n.

Sharkovskii's theorem

Li and Yorke were not the first. In 1964 the Ukrainian mathematician Oleksandr Sharkovskii had proved a far more complete result, published in Russian in the *Ukrainian Mathematical Journal* and unknown in the West for a decade. Order the positive integers as

$$3,\ 5,\ 7,\ 9,\ \ldots,\ \ 2\cdot 3,\ 2\cdot 5,\ \ldots,\ \ 2^2\cdot 3,\ 2^2\cdot 5,\ \ldots,\ \ \ldots,\ \ 2^3,\ 2^2,\ 2,\ 1:$$

first the odd numbers from 3 upwards, then twice the odd numbers, then four times the odd numbers, and so on, ending with the powers of 2 in decreasing order. Sharkovskii's theorem says that if a continuous function on an interval has a cycle of length m, then it has cycles of every length that comes after m in this list. Since 3 comes first, period 3 implies every period. And the theorem is sharp: for every m there is a function with a cycle of length m and no cycles of lengths earlier in the list. The powers of 2 at the end are exactly the period-doubling cascade of the logistic map in Section 8.7, and the 3-cycle appears there at $k = 1 + \sqrt{8}$.

The tent map and the logistic map

Substitute $x = 14y - 7$, which maps $[0,1]$ onto $[-7,7]$. A short calculation shows that f becomes the *tent map*

$$T(y) = \begin{cases} 2y & 0 \le y \le \frac{1}{2}, \\ 2 - 2y & \frac{1}{2} \le y \le 1, \end{cases}$$

and the further substitution $u = \sin^2(\pi y/2)$ turns the tent map into the logistic map $u \mapsto 4u(1-u)$. So the function in this question is the chaotic logistic map of Section 8.7, straightened out. Its eight solutions of $f^3(x) = x$ correspond to the eight solutions of $\sin^2(8\theta) = \sin^2\theta$ in $[0, \frac{\pi}{2}]$.

Try this next

1. **A 4-cycle.** Use the covering diagram to predict where a 4-cycle of f should be (three steps in I_2, one in I_1). Then find it exactly. (*Hint:* the cycle has three positive points and one negative point; write down the four linear equations.)

2. **The subtlety.** In the proof that period 3 implies period n, explain why the point x found cannot be the point 1 (or any point of the original 3-cycle) when $n \ge 4$.

3. **Period 5 without period 3.** Sketch the graph of a continuous function on $[1,5]$ with $g(1) = 3$, $g(3) = 4$, $g(4) = 2$, $g(2) = 5$, $g(5) = 1$, joining these points with straight lines. Draw its covering diagram for the intervals $[1,2]$, $[2,3]$, $[3,4]$, $[4,5]$. Can you find a loop of length 3? (Sharkovskii's theorem says there is no 3-cycle.)

4. **The tent map.** Check that $x = 14y - 7$ turns f into the tent map. Then show that the tent map doubles y written in binary, flipping all the digits whenever the leading digit is 1, and use this to find its 3-cycles.

Further reading

- **Tien-Yien Li & James Yorke, "Period three implies chaos"**, *American Mathematical Monthly* 82 (1975). Short, and readable with the intermediate value theorem.
- **Robert Devaney, *A First Course in Chaotic Dynamical Systems*** (CRC Press). Written for undergraduates; covers the tent map, the logistic map and Sharkovskii's theorem with full proofs.
- **Keith Burns & Boris Hasselblatt, "The Sharkovsky theorem: a natural direct proof"**, *American Mathematical Monthly* 118 (2011). A modern proof using covering diagrams.

The full guided solution is in Section 8.8:
https://beyondstep.keithmcnulty.org/ch08/period-three.html